

Climate and Forecasting Aggregation (CFA) file preparation for the CANARI climate model ensemble

Jonny Williams*, David Hassell, Bryan Lawrence
NCAS-CMS

May 28, 2026

Contents

1	Introduction	3
2	Workflow	3
2.1	Aggregation with cf Python	4
2.2	Workflow diagram	4
2.3	Seed	6
2.4	Grow	7
2.4.1	Note on historical diagnostic sets	7
2.5	Arrange	7
2.5.1	JDMA batch IDs	7
2.5.2	JDMA ‘external’ IDs	7
3	Example usage and validation	12
3.1	One day of 3-hourly, 1.5 m air temperature	12
3.2	Extension across batches	12
4	Conclusions	15
A	Benchmarking	15
A.1	Seed	15
A.2	Grow	16
A.3	Arrange	16
A.4	Comparison of time taken in seed, grow and arrange steps	16
B	Runtime instructions	16
B.1	Seed	18
B.2	Grow	18
B.3	Arrange	19
C	Priority variables	19
C.1	Background	19
C.2	Algorithm	19
C.3	Discovery and generation of single-variable files	25
C.3.1	Retrieval	25
C.3.2	Extraction to yearly	27

*jonny.williams@ncas.ac.uk

C.4 Validation of corrupt priority-variable files	28
---	----

JDMA batch numbers

```
scil.jasmin.ac.uk:reinhard$ jdma batch -f workspace -w canari | grep u-cv575 | cat -n
  1  3203 reinhard      canari      u-cv575/19500101 elastictape  2023-03-22 15:44 ON_STORAGE
  2  3230 reinhard      canari      u-cv575/19500401 elastictape  2023-03-23 15:44 ON_STORAGE
  3  3231 reinhard      canari      u-cv575/19500701 elastictape  2023-03-23 15:54 ON_STORAGE
  4  3232 reinhard      canari      u-cv575/19501001 elastictape  2023-03-23 16:05 ON_STORAGE
  5  3240 reinhard      canari      u-cv575/19510101 elastictape  2023-03-23 23:49 ON_STORAGE
  6  3255 reinhard      canari      u-cv575/19510401 elastictape  2023-03-24 03:10 ON_STORAGE
  7  3263 reinhard      canari      u-cv575/19510701 elastictape  2023-03-24 08:51 ON_STORAGE
  8  3270 reinhard      canari      u-cv575/19511001 elastictape  2023-03-24 11:42 ON_STORAGE
  9  3279 reinhard      canari      u-cv575/19520101 elastictape  2023-03-24 16:34 ON_STORAGE
 10  3304 reinhard      canari      u-cv575/19520401 elastictape  2023-03-24 20:04 ON_STORAGE
 11  3310 reinhard      canari      u-cv575/19520701 elastictape  2023-03-24 22:45 ON_STORAGE
 12  3400 reinhard      canari      u-cv575/19521001 elastictape  2023-03-27 08:17 ON_STORAGE
 13  3401 reinhard      canari      u-cv575/19530101 elastictape  2023-03-27 08:22 ON_STORAGE
 14  3402 reinhard      canari      u-cv575/19530401 elastictape  2023-03-27 08:27 ON_STORAGE
 15  3403 reinhard      canari      u-cv575/19530701 elastictape  2023-03-27 08:32 ON_STORAGE
 16  3404 reinhard      canari      u-cv575/19531001 elastictape  2023-03-27 08:42 ON_STORAGE
 17  3405 reinhard      canari      u-cv575/19540101 elastictape  2023-03-27 08:48 ON_STORAGE
```

Figure 1: Example of JDMA batch indexing from the CANARI project pages.

1 Introduction

This document details the processes by which data from the CANARI project – large ensembles of simulations using the HadGEM3-GC3.1 climate model [Williams et al., 2018] – has been aggregated after its ingestion into the Joint-storage Data Migration App (JDMA) archive¹. The JDMA archive is hosted on the JASMIN platform hosted by the Centre for Environmental Data Analysis, CEDA².

The CANARI ensemble consists of two, 40-member ensembles:

1. Historical simulations running from 1850-2015.
2. SSP3-7.0 future scenarios, 2015-2100.

SSP stands for Shared Socioeconomic Pathway. These are a family of potential future greenhouse gas and land-use pathways which are standardised and extensively used in the climate science community, more information can be found in IPCC [2023].

Within the respective ensembles, each member differs only in its initial conditions, which are generated using a hybrid ‘macro-micro’ initialisation method [Schiemann et al., 2026].

Each simulation generates data in three month ‘batches’ and the format of the batches’ contents are described as, e.g., `runid/cyclepoint`, where `runid` takes the form `u-ab123` and `cyclepoint` is in the ISO-8601 ‘datetime’ standard, e.g. `19500101T0000Z` [International Organization for Standardization, 2019]. Each batch is then given an integer index which describes its ‘location’ on the JDMA archive. These values are given on the NCAS-CMS GitHub pages³ and an example is given in Figure 1. The data is held in monthly files and so each batch contains three sets of data in a common format.

2 Workflow

The workflow discussed here assumes that the diagnostic content of each simulation is the same. In almost every respect this is the case, except for four variables in the atmospheric output in the historical simulations; this will be discussed in § 2.4.1. This assumption significantly reduces the time taken to generate the ‘seed’ files described below which make up the vast majority of

¹<https://help.jasmin.ac.uk/docs/short-term-project-storage/jdma/>

²<https://www.ceda.ac.uk>

³<https://ncas-cms.github.io/canari>

ID	Historical	SSP3-7.0	ID	Historical	SSP3-7.0
1	u-cv575 *	u-de814	21	u-da179	u-df933
2	u-cv625 *	u-de436	22	u-da190	u-df934
3	u-cw345 *	u-de724	23	u-da191	u-df935
4	u-cw356 *	u-de815	24	u-da192	u-df936
5	u-cv827 *	u-df220	25	u-da193	u-df937
6	u-cv976 *	u-de830	26	u-db291	u-dh412
7	u-cz547 *	u-de831	27	u-db301	u-dh413
8	u-cy436 *	u-de832	28	u-db303	u-dh415
9	u-cw342 *	u-de850	29	u-db304	u-dh416
10	u-cw343 *	u-de851	30	u-db305	u-dh417
11	u-cy375 *	u-de934	31	u-cz475	u-di511
12	u-cy376 *	u-de937	32	u-cz568	u-di512
13	u-cy537	u-de938	33	u-cz647	u-di513
14	u-cy811	u-de939	34	u-cz648	u-di514
15	u-cy866	u-de940	35	u-cz649	u-di515
16	u-cy873	u-df299	36	u-dd436	u-di703
17	u-cy877	u-df300	37	u-dd438	u-di704
18	u-cy879	u-df301	38	u-dd439	u-di705
19	u-cy880	u-df302	39	u-dd441	u-di706
20	u-cy881	u-df303	40	u-dd442	u-di707

Table 1: Ensemble Member RunIDs for Control and SSP3-7.0 scenarios. The simulations marked with asterisks have a slightly different output variable set and the ensemble members used as ‘seeds’ are highlighted; see § 2.4.

the processing time. Table 1 shows the simulation identifiers for the historical and SSP3-7.0 simulations. Their format is standard for climate models run with the Rose and Cylc [Oliver et al., 2018] workflow tools. Rose is ‘a framework for managing and running meteorological suites’⁴ and Cylc is ‘a general purpose workflow engine with a particular gift for cycling’⁵.

2.1 Aggregation with cf Python

CF aggregation is a new feature in the CF conventions, having been available since December 2025 [Eaton et al., 2025]. Their usage in this particular application is in the representation of a large database of climate model data – $\mathcal{O}(\text{PB})$ – in a format which is portable and easily reproducible – $\mathcal{O}(100\text{MB})$ or less. The workflow must also be achievable in a timescale which enables easy reruns and big fixes, $\mathcal{O}(1 \text{ day})$ or less.

2.2 Workflow diagram

After choosing the scenario and seed simulation ID, the workflow is made up of three stages; seed, grow and arrange. This is illustrated in Figure 2. The individual algorithms for the seed, grow and arrange steps are shown in Figure 4, 5 and 7 respectively.

⁴<https://www.metoffice.gov.uk/research/approach/modelling-systems/rose>

⁵<https://cylc.github.io>

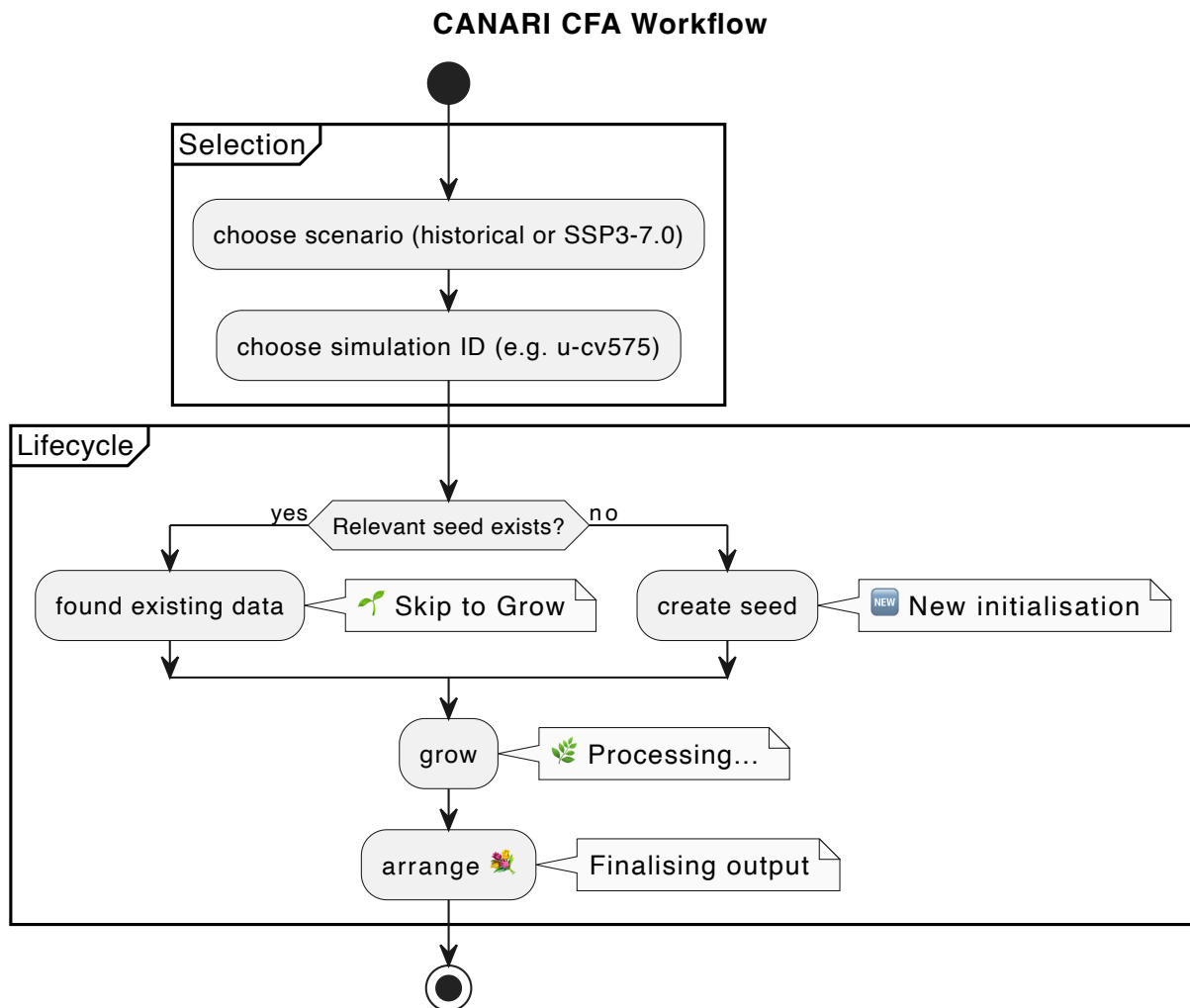


Figure 2: Workflow diagram of the whole process of generating the CFA files, including choosing the scenario and simulation ID. Figure prepared using PlantUML.

```
fragment_unique_values = "195001" ;

m01s00i002 = _ ;

fragment_map_1 =
{120},
{85},
{324},
{432} ;

fragment_uris =
"cv575a_1_6hr_u_pt_cordex__195001-195001.nc" ;
```

Figure 3: `fragment_unique_values` and `fragment_unique_values` variables in the historical simulations' seed CFA file.

2.3 Seed

The seed stage requires one complete month (only) of data on locally accessible disk. It is this month which is interrogated by the Python script `seed.py` and generates a seed file which is then 'grown' – § 2.4 – and 'arranged' – § 2.5 – to give the final CFA file for the complete run length of the simulation (see page 3).

For example, the files which are used in the creation of the seed file for the historical ensemble are of the form `cv575{a,o,i}_1*195001*.nc`, where `{a,o,i}` uses the standard Linux wildcard syntax for 'a or o or i' to refer to the atmosphere, ocean and sea ice datasets respectively. Going forward we use the term 'realm' to refer to this distinction. This in turn assumes a consistent naming convention of the post processed ensemble data, and we assume this to be the case throughout the ensembles.

Of primary importance for the understanding of later sections of this document are the variables which describe the following:

- File names
 - These are called `fragment_uris` in the language of the CF conventions⁶ where `uris` are Uniform Resource Identifiers.
- JDMA batch numbers
 - These are called `fragment_unique_values` in the CF conventions.

Figure 3 shows a screenshot of a section of the `ncdump` output from the resulting seed CFA file. The `fragment_unique_values` and `fragment_uris` variables are highlighted.

Note that in the seed file, the `fragment_unique_values` string is set to the value of `year+month` strings in the `seed.py` file where 01 is January etc. This convention is also used in the `grow.py` described in § 2.4.

⁶<https://cfconventions.org>

The seed Python file also removes the NetCDF `name` attribute, which by default points to the original creation location on disk when the simulation was first run and is hence irrelevant – and potentially misleading – once the data has undergone any processing. The `seed.py` file must be run separately for each realm as is the case for the ‘grow’ and ‘arrange’ steps too.

Finally, Figure 4 shows a visual representation of the algorithm used in `seed.py`.

2.4 Grow

Figure 5 shows the PlantUML visualisation of the `grow.py` script. The majority of the code in this step of the overall workflow is concerned with the formatting of the date-time strings which, as part of the filenames – i.e. `fragment_uris` – are grown, month by month, up to the end of the simulation’s runtime (historical, 65 years; SSP3-7.0 85 years).

Figure 6 shows the first 14 months for the `fragment_uris` variable⁷. The equivalent `fragment_unique_values` variables are simply a reproduction – to the same size as the `fragment_uris` – of that from the seeding step, i.e., placeholders of the form 195001 and 201501 for the historical and SSP3-7.0 ensembles respectively. The `arrange.py` edits these ‘dummy’ values and is described in § 2.5.

The growth step is only required to be run for the ‘seed’ simulations, i.e. the highlighted ones in Table 1. The arrangement step creates the final required files for the entire ensemble.

2.4.1 Note on historical diagnostic sets

For the historical ensemble, members 1-13 have slightly different variable sets⁸ to the remainder. Therefore, members 2-12 were seeded from member 1, `u-dv575`, and 14-40 were seeded from member 13, `u-cy537`. All SSP3-7.0 members were seeded from its first member, `u-de814`.

2.5 Arrange

2.5.1 JDMA batch IDs

Figure 2.5 shows the algorithm for the ‘arrangement’ step.

When the CANARI ensemble was run, the JDMA batch numbers associated with each three month cycle were manually documented – see page 1 – and these values are used as ‘lookup tables’ to find the relevant batch number from the datetime string in the `fragment_uri` values. An important step here is to convert the monthly – 195001, 195002 – datetime strings into the correct three-monthly cycle points. The need for this is illustrated in Figure 1 where the batches are names 195001, 195004, etc. This is the ‘quarter map’ step shown in Figure 7.

The results of the arrangement step is shown in Figure 8, which shows the `fragment_unique_values` generated from the `arrange.py` script for the equivalent information shown in the CANARI online documentation in Figure 9.

2.5.2 JDMA ‘external’ IDs

Ongoing work will see the data on the JDMA archive moved to the Near-Line Data Store (NLDS).⁹ To assist in this process, the elastic tape ‘external IDs’ are also required. These numbers are found from the `jdma batch` command as shown in Listing 1):

⁷Note that this is the first of these, the *second* is named `fragment_uris_1` and so forth.

⁸see <https://ncas-cms.github.io/canari/>

⁹<https://help.jasmin.ac.uk/docs/short-term-project-storage/nlds/>

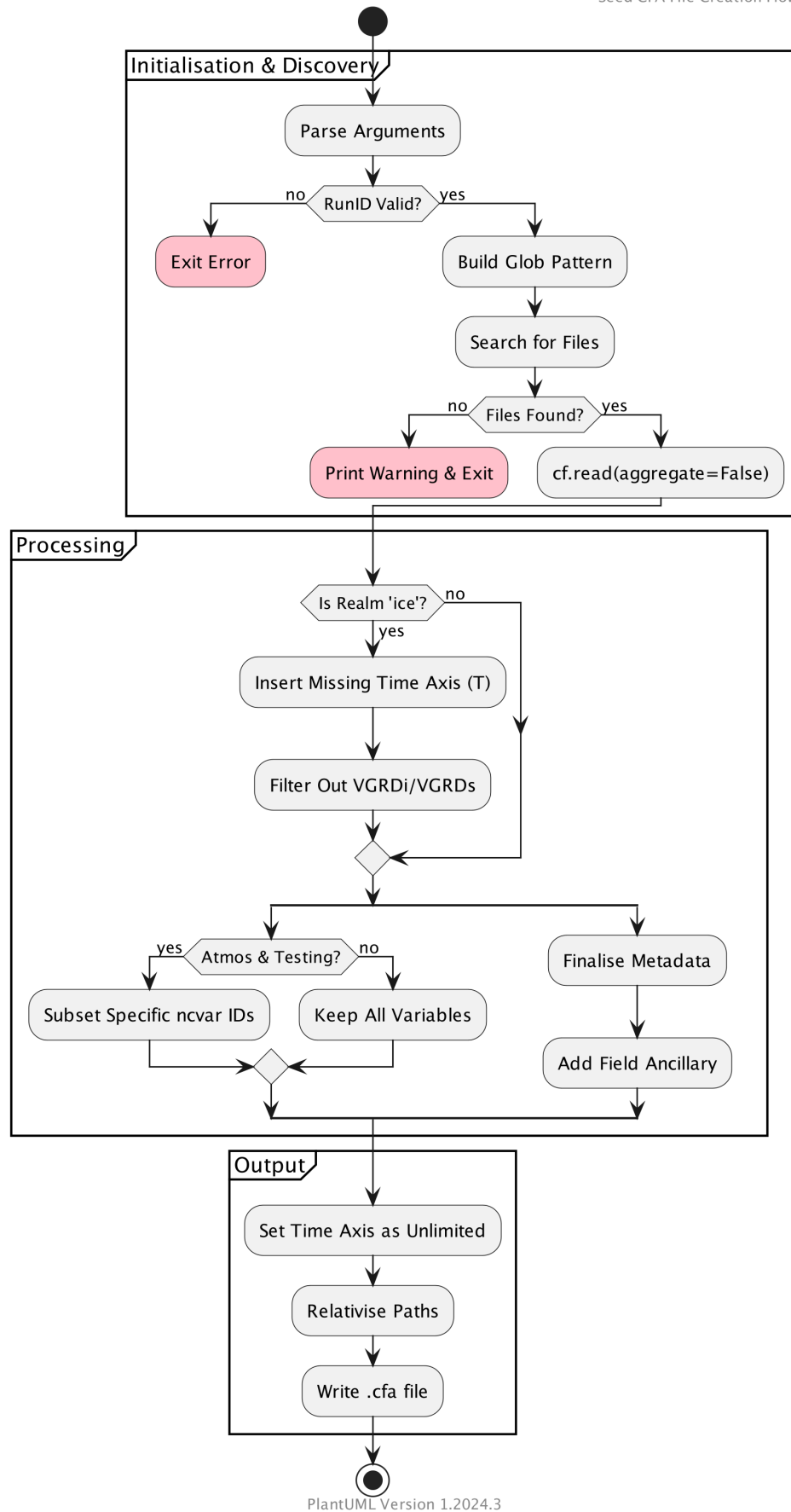


Figure 4: Algorithm for the `seed.py` file, prepared with PlantUML [Roques, 2026].

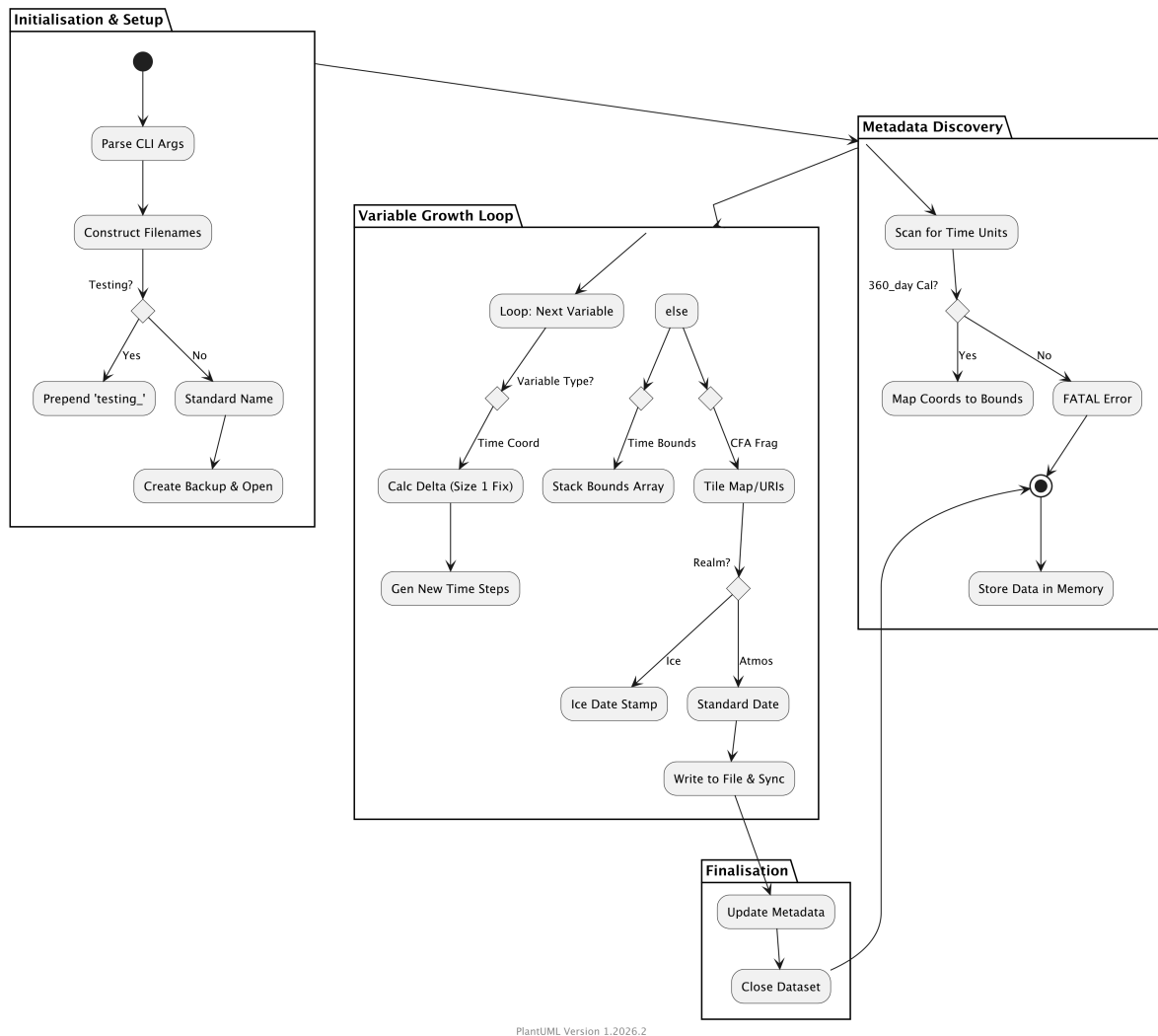


Figure 5: Algorithm for the `grow.py` file.

```

fragment_uris =
"de814a_1_6hr_u_pt_cordex__201501-201501.nc",
"de814a_1_6hr_u_pt_cordex__201502-201502.nc",
"de814a_1_6hr_u_pt_cordex__201503-201503.nc",
"de814a_1_6hr_u_pt_cordex__201504-201504.nc",
"de814a_1_6hr_u_pt_cordex__201505-201505.nc",
"de814a_1_6hr_u_pt_cordex__201506-201506.nc",
"de814a_1_6hr_u_pt_cordex__201507-201507.nc",
"de814a_1_6hr_u_pt_cordex__201508-201508.nc",
"de814a_1_6hr_u_pt_cordex__201509-201509.nc",
"de814a_1_6hr_u_pt_cordex__201510-201510.nc",
"de814a_1_6hr_u_pt_cordex__201511-201511.nc",
"de814a_1_6hr_u_pt_cordex__201512-201512.nc",
"de814a_1_6hr_u_pt_cordex__201601-201601.nc",
"de814a_1_6hr_u_pt_cordex__201602-201602.nc",

```

Figure 6: Example selection (first 14 months) `fragment_uris` of the grown atmosphere CFA file for the first SSP3-7.0 ensemble member.

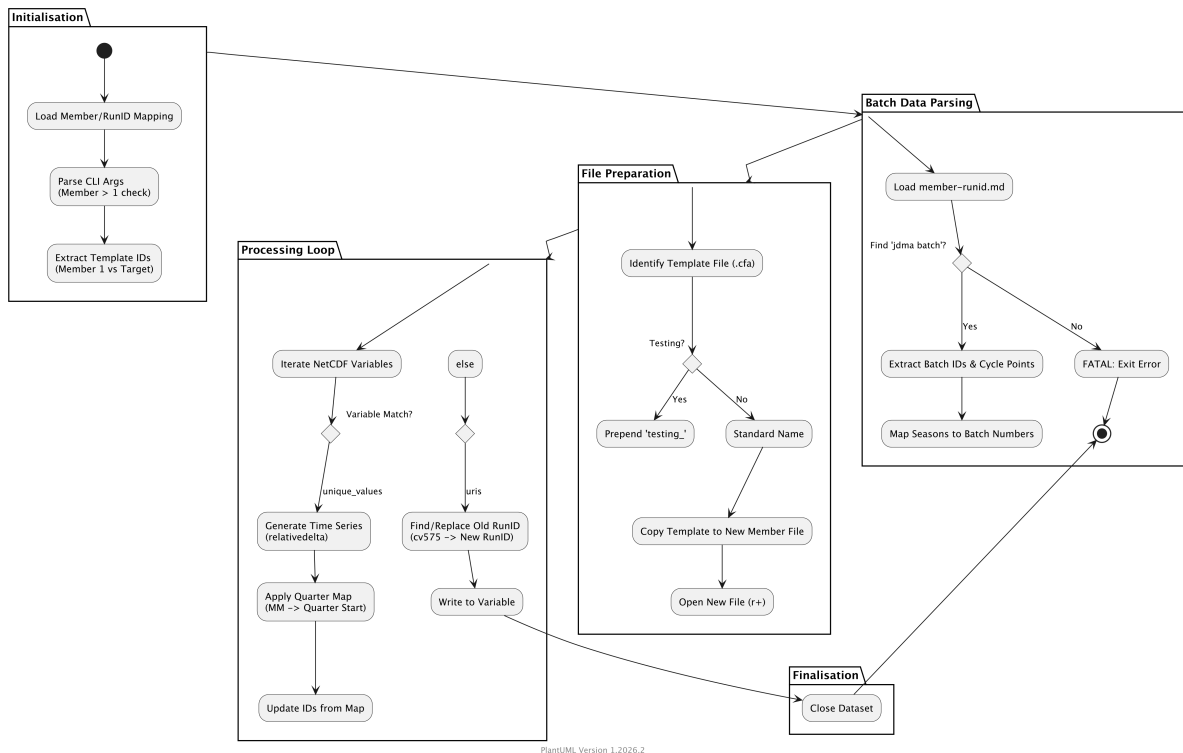


Figure 7: Algorithm for the `arrange.py` file.

```
fragment_unique_values = "18164", "18164", "18164", "18171", "18171",
"18171", "18178", "18178", "18178", "18185", "18185", "18185", "18266",
"18266", "18266", "18313", "18313", "18313", "18335", "18335", "18335",
"18361", "18361", "18361", "18372", "18372", "18372", "18377", "18377",
"18377", "18392", "18392", "18392", "18417", "18417", "18417", "18438",
"18438", "18438", "18445", "18445", "18445", "18479", "18479", "18479",
"18502", "18502", "18502", "18676", "18676", "18676", "18684", "18684",
"18684", "18686", "18686", "18686", "18689", "18689", "18689", "18694",
"18694", "18694", "18700", "18700", "18700", "18705", "18705", "18705",
"18720", "18720", "18720", "18732", "18732", "18732", "18759", "18759",
"18759", "18774", "18774", "18774", "18792", "18792", "18792", "18816",
"18816", "18816", "18826", "18826", "18826", "18838", "18838", "18838",
```

Figure 8: The `fragment_unique_values` variable for the first several years of the simulation `u-de814`.

```
$ jdma batch -f workspace -w canari | grep u-de814 | grep -viE "deleted|failed" | cat -n
 1  18164 reinhard canari u-de814/20150101 elastictape 2024-03-25 10:28 ON_STORAGE
 2  18171 reinhard canari u-de814/20150401 elastictape 2024-03-25 10:33 ON_STORAGE
 3  18178 reinhard canari u-de814/20150701 elastictape 2024-03-25 10:38 ON_STORAGE
 4  18185 reinhard canari u-de814/20151001 elastictape 2024-03-25 10:43 ON_STORAGE
 5  18266 reinhard canari u-de814/20160101 elastictape 2024-03-26 04:08 ON_STORAGE
 6  18313 reinhard canari u-de814/20160401 elastictape 2024-03-26 15:15 ON_STORAGE
 7  18335 reinhard canari u-de814/20160701 elastictape 2024-03-27 04:37 ON_STORAGE
 8  18361 reinhard canari u-de814/20161001 elastictape 2024-03-27 12:49 ON_STORAGE
 9  18372 reinhard canari u-de814/20170101 elastictape 2024-03-27 15:35 ON_STORAGE
10  18377 reinhard canari u-de814/20170401 elastictape 2024-03-27 18:47 ON_STORAGE
11  18392 reinhard canari u-de814/20170701 elastictape 2024-03-28 03:28 ON_STORAGE
12  18417 reinhard canari u-de814/20171001 elastictape 2024-03-28 12:29 ON_STORAGE
13  18438 reinhard canari u-de814/20180101 elastictape 2024-03-28 16:31 ON_STORAGE
14  18445 reinhard canari u-de814/20180401 elastictape 2024-03-28 19:28 ON_STORAGE
15  18479 reinhard canari u-de814/20180701 elastictape 2024-03-29 23:20 ON_STORAGE
16  18502 reinhard canari u-de814/20181001 elastictape 2024-03-30 07:07 ON_STORAGE
17  18676 reinhard canari u-de814/20190101 elastictape 2024-04-02 15:27 ON_STORAGE
18  18684 reinhard canari u-de814/20190401 elastictape 2024-04-02 20:23 ON_STORAGE
19  18686 reinhard canari u-de814/20190701 elastictape 2024-04-02 22:24 ON_STORAGE
20  18689 reinhard canari u-de814/20191001 elastictape 2024-04-02 23:59 ON_STORAGE
21  18694 reinhard canari u-de814/20200101 elastictape 2024-04-03 00:54 ON_STORAGE
22  18700 reinhard canari u-de814/20200401 elastictape 2024-04-03 01:50 ON_STORAGE
23  18705 reinhard canari u-de814/20200701 elastictape 2024-04-03 02:50 ON_STORAGE
24  18720 reinhard canari u-de814/20201001 elastictape 2024-04-03 06:26 ON_STORAGE
25  18732 reinhard canari u-de814/20210101 elastictape 2024-04-03 14:49 ON_STORAGE
26  18759 reinhard canari u-de814/20210401 elastictape 2024-04-04 07:01 ON_STORAGE
27  18774 reinhard canari u-de814/20210701 elastictape 2024-04-04 14:43 ON_STORAGE
28  18792 reinhard canari u-de814/20211001 elastictape 2024-04-04 22:04 ON_STORAGE
29  18816 reinhard canari u-de814/20220101 elastictape 2024-04-05 06:25 ON_STORAGE
30  18826 reinhard canari u-de814/20220401 elastictape 2024-04-05 10:11 ON_STORAGE
31  18838 reinhard canari u-de814/20220701 elastictape 2024-04-05 14:47 ON_STORAGE
```

Figure 9: The JDMA batch numbers variable for the first several years of the simulation `u-de814`. The period shown is the same as shown in Figure 8.

Listing 1: `jdma batch` output

```
jdma batch 3203
  Batch for user: pug123
  Batch id      : 3203 <--- Needed in CFA files!
  Workspace     : canari
  Batch label   : u-cv575/19500101T0000Z
  Ex. storage   : elastictape
  Date          : 2023-03-22 15:44
  External id   : 33076 <--- Needed in CFA files!
  Stage        : ON_STORAGE
```

The final format of the `fragment_unique_values` strings is, e.g., `3203_33076`, i.e. `[batch_ID]_[external_ID]`.

3 Example usage and validation

Here we check that the output CFA files do indeed match the expected structure imposed on them by the method outlined above.

3.1 One day of 3-hourly, 1.5 m air temperature

Figure 10 shows code from a Jupyter notebook where the resulting CFA file for member 1 of the historical ensemble is opened using `cf.read`. After this, the 1.5 m temperature is read. This diagnostic appears many times in the output (on different time and spatial domains) and so we have then further concentrated on 3-hourly values, resulting in a single field. We then choose a random day in the 1950-2014 time series and examine the properties of the resulting 8 fields.

For this example, the date is 12/11/2004 and this is correctly identified with the file `db301a_13_3hr_pt__200411-200411.nc`. The field's ancillary data are the JDMA batches which in this case are all the same since we are only looking at one day.

The validation stage is to interrogate the data in the CANARI GitHub repository, which is itself the raw output of the `jdma batch` command, an example of which is shown in Figure 1. The final lines of output shown in Figure 10 show that on the 220th line of the relevant markdown file, the JDMA batch is 17007, matching the ancillary data values from the CFA file. Additionally, `jdma batch 17007|grep -i external` gives `External id : 54049` as expected.

3.2 Extension across batches

Here we examine an analogous situation to that in § 3.1 but where the period in question extends over multiple batches, see Figure 11.

The code shown in Figure 11 is slightly modified compared to Figure 10 in that the initial month selection is limited to January-September (inclusive) to ensure that when a 3-month 'time delta' is added to the randomly chosen date, the year does not need to be changed to reflect this. This is purely for illustrative convenience and is not a limitation of the code.

Figure 11 gives the randomly assigned initial date as 21/1/1998 and extends to 21/4/1998 giving the following list of files (the highlighted file indicates its presence in a different JDMA batch to the others):

1. `db301a_13_3hr_pt__199801-199801.nc`
2. `db301a_13_3hr_pt__199802-199802.nc`
3. `db301a_13_3hr_pt__199803-199803.nc`

```

1 f = cf.read(
2     glob.glob("./CF-1.13_grown_CANARI_27_db301_atmos_65years_batches.cfa"),
3 )
4
5 y = np.random.randint(1950, 2014)
6 m = np.random.randint(1, 12)
7 d = np.random.randint(1, 30)
8 print(f"The date is {y:04d}-{m:02d}-{d:02d}")
9
10 target_field = f.select_by_property(
11     standard_name="surface_temperature", interval_operation="3 h"
12 )[0].subspace(T=cf.wi(cf.dt(y, m, d, 0, 0), cf.dt(y, m, d, 23, 59)))
13
14 pprint("File name or names...")
15 pprint(target_field.data.get_filenames())
16 pprint("The unique ancillary batch or batches are...")
17 np.unique(
18     target_field.constructs.filter(
19         filter_by_property={"long_name": re.compile(".*jdma.*", re.I)}
20     )
21     .value()
22     .array
23 ).tolist()

```

```

The date is 2004-11-12
'File name or names...'
{'db301a_13_3hr_pt_200411-200411.nc'}
'The unique ancillary batch or batches are...'
['17007_54049']

```

```

1 # for _m in [m, m + 3]:
2
3 m_formatted = f"{m:02d}"
4 search_pattern = str(y) + quarter_map[m_formatted]
5
6 for file in glob.glob("/home/users/jonnyhtw/canari/docs/hist2/*db301*"):
7     for line in open(file):
8         if search_pattern in line:
9             print(line)

```

220 17007 jrobson canari u-db301/20041001 elastictape 2024-02-23 16:55 ON_STORAGE

Figure 10: Validation of one day of 3-hourly 1.5 m air temperature output; comparison of CFA file resulting from this work and the raw JDMA batch information.

```

1 f = cf.read(
2 |   glob.glob("./CF-1.13_grown_CANARI_27_db301_atmos_65years_batches.cfa"),
3 | )
4
5 y = np.random.randint(1950, 2014)
6 m = np.random.randint(1, 9)
7 d = np.random.randint(1, 30)
8 print(f"The date is {y:04d}-{m:02d}-{d:02d}")
9
10 target_field = f.select_by_property(
11 |   standard_name="surface_temperature", interval_operation="3 h"
12 | )[0].subspace(T=cf.wi(cf.dt(y, m, d, 0, 0), cf.dt(y, m + 3, d, 23, 59)))
13
14 pprint("File name or names...")
15 pprint(target_field.data.get_filenames())
16 pprint("The unique ancillary batch or batches are...")
17 np.unique(
18 |   target_field.constructs.filter(
19 |     filter_by_property={"long_name": re.compile(".*jdma.*", re.I)}
20 |   )
21 |   .value()
22 |   .array
23 | ).tolist()

```

```

The date is 1998-01-21
'File name or names...'
{'db301a_13_3hr_pt__199801-199801.nc',
 'db301a_13_3hr_pt__199802-199802.nc',
 'db301a_13_3hr_pt__199803-199803.nc',
 'db301a_13_3hr_pt__199804-199804.nc'}
'The unique ancillary batch or batches are...'
['16552_52832', '16557_52853']

```

```

1 for _m in [m, m + 3]:
2
3     m_formatted = f"{_m:02d}"
4     search_pattern = str(y) + quarter_map[m_formatted]
5
6     for file in glob.glob("/home/users/jonnyhtw/canari/docs/hist2/*db301*"):
7         for line in open(file):
8             if search_pattern in line:
9                 print(line)

```

193	16552 jrobson	canari	u-db301/19980101 elasticsearch	2024-02-17 01:24	ON_STORAGE
194	16557 jrobson	canari	u-db301/19980401 elasticsearch	2024-02-17 04:29	ON_STORAGE

Figure 11: As for Figure 10 but over a period of 3 months, i.e. traversing two JDMA batches.

4. db301a_13_3hr_pt__199804-199804.nc

The unique JDMA batch numbers are given as 16552 and 16557. Again these agree with the ‘raw’ `jdma batch` output.

For the external IDs, once again they agree (Figure 11).

- `jdma batch 16552|grep -i external External id : 52832`
- `jdma batch 16557|grep -i external External id : 52853`

4 Conclusions

This work has expanded on previous work to catalogue the variables in the CANARI historical and SSP3-7.0 scenario ensembles. This has been done using the aggregation capabilities of the `cf` Python package [Hassell et al., 2017], termed ‘CFA’.

This process involves three discrete steps; seeding, growth and arrangement. The first of these generates the template (i.e. ‘seed’) for the first month of a reference simulation, the second grows it out to the length of the simulation inferring the already-standardised filenames, and the third assigns the correct tape archive ‘batch’ numbers thus enabling later file discovery (‘arrangement’).

The work has been version controlled using Git and is broadly applicable to similarly-standardised datasets of arbitrary size.

The largest time resource is taken up by the primary seeding step. This can be seen in Appendix A which shows that the generation of the seed file for the atmosphere realm of the historical scenario takes approximately 1 hour, the growth step take 16 minutes and arrangement 6 seconds.

That said, the seeding only needs to be run once for each data instance; e.g. atmosphere data for a particular climate scenario. In this case there was a small change to the diagnostic set in ensemble members 13-40 which necessitated the generation of a different seed, but this only adds $\mathcal{O}(1 \text{ hour})$ to the overall computation time since a new seed file must be created.

These scripts – although predicated on the CANARI ensemble and pre-formatted filenames – could straightforwardly be edited and used by other large datasets.

Appendix A Benchmarking

This section compares the time taken in each of the seed, grow and arrange steps for the atmosphere realm of `u-cy537`, the 13th member of the historical ensemble; see Table 1.

A.1 Seed

Listing 2

Listing 2: `seed.py`

```
time python seed.py --runid    cy537\  
                    --member   13\  
                    --realm    atmos\  
                    --data_path $HOME/archive/CANARI/u-cy537/19500101T0000Z/\br/>                    --startyear 1950
```

Table 2: Timings for arrange, historical, atmosphere, using the output from the `time` Linux command. The ‘real’ time is the ‘Elapsed real time (in seconds)’, ‘user’ is ‘Total number of CPU-seconds that the process spent in user mode.’ and ‘sys’ is ‘Total number of CPU-seconds that the process spent in kernel mode.’ where the definitions are from the `man time` Linux command.

Measure	time
real	59 m 15 s
user	55 m 18 s
sys	2 m 37 s

A.2 Grow

As for § A.1 for the growth step. Code usage in Listing 3.

Listing 3: `grow.py`

```
time python grow.py --runid cy537\  
                    --member 13\  
                    --realm  atmos\  
                    --n_years 65
```

Table 3: Timings for grow, historical, atmosphere, using the output from the `time` Linux command. The ‘real’ time is the ‘Elapsed real time (in seconds)’, ‘user’ is ‘Total number of CPU-seconds that the process spent in user mode.’ and ‘sys’ is ‘Total number of CPU-seconds that the process spent in kernel mode.’ where the definitions are from the `man time` Linux command.

Measure	time
real	16 m 16 s
user	15 m 16 s
sys	0 m 27 s

A.3 Arrange

As for § A.1 for the arrange step. Code usage shown in Listing 4.

Listing 4: `arrange.py`

```
time python arrange.py --member 13\  
                      --realm  atmos\  
                      --n_years 65\  
                      --scenario hist2
```

A.4 Comparison of time taken in seed, grow and arrange steps

Figure 12 shows the ‘real’ time taken as shown in Tables 2, 3 and 4.

Appendix B Runtime instructions

This section shows the command line ‘doc string’ help available for the seed, grow and arrange steps.

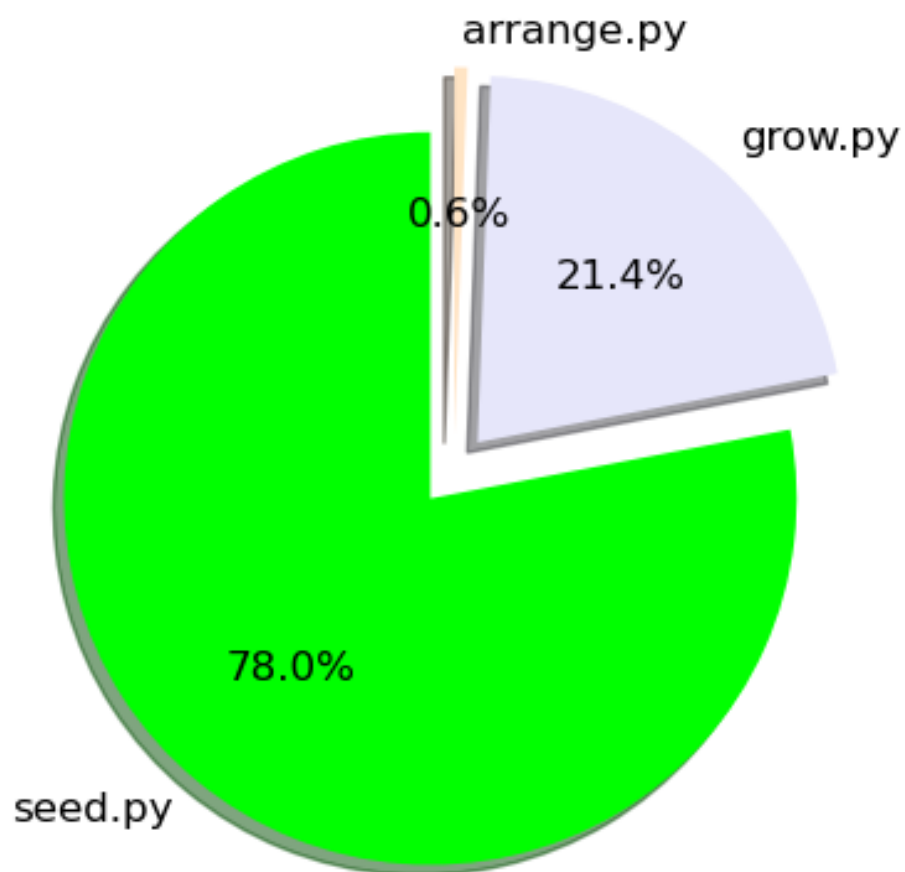


Figure 12: Pie graph representation of the 'real' times from Tables 2, 3 and 4.

Table 4: Timings for arrange, historical, atmosphere, using the output from the `time` Linux command. The ‘real’ time is the ‘Elapsed real time (in seconds)’, ‘user’ is ‘Total number of CPU-seconds that the process spent in user mode.’ and ‘sys’ is ‘Total number of CPU-seconds that the process spent in kernel mode.’ where the definitions are from the `man time` Linux command.

Measure	time
real	0 m 26 s
user	0 m 25 s
sys	0 m 2 s

B.1 Seed

Listing 5.

Listing 5: Output of `python seed.py -h`

```
usage: seed.py [-h] --runid RUNID --realm {atmos,ocean,ice} --member MEMBER [--testing
{yes,no}] --startyear STARTYEAR
               --data_path DATA_PATH

Initializes CFA files from source data.

options:
  -h, --help                show this help message and exit
  --runid RUNID              Run ID in format letter-letter-number-number-number
  --realm {atmos,ocean,ice} The climate realm
  --member, -m MEMBER       The ensemble member number
  --testing, -t {yes,no}    Is this a test run? (default: no)
  --startyear STARTYEAR     Starting year of the simulation.
  --data_path, -d DATA_PATH Path to the source data directory

(cfa_env) sci-vm-01|Mon Mar 23|16:51:12 GMT|cfa_scripts|[main]>
```

B.2 Grow

Listing 6.

Listing 6: Output of `python grow.py -h`

```
usage: grow.py [-h] --runid RUNID --realm {atmos,ocean,ice} --member MEMBER [--testing
{yes,no}] --n_years N_YEARS

Appends data to existing CFA files.

options:
  -h, --help                show this help message and exit
  --runid RUNID              Run ID in format letter-letter-number-number-number
  --realm {atmos,ocean,ice} The climate realm
  --member, -m MEMBER       The ensemble member number
  --testing, -t {yes,no}    Is this a test run? (default: no)
  --n_years, -ny N_YEARS
```

```
(cfa_env) sci-vm-01|Mon Mar 23|16:52:48 GMT|cfa_scripts|[main]>
```

B.3 Arrange

Listing 7.

Listing 7: Output of python arrange.py -h

```
usage: arrange.py [-h] --member MEMBER --realm {atmos,ocean,ice} [--testing {yes,no}]
               --n_years N_YEARS --scenario {hist2,ssp370}

Adds in JDMA batch numbers.

options:
  -h, --help                show this help message and exit
  --member, -m MEMBER      Ensemble member.
  --realm {atmos,ocean,ice}
  --testing, -t {yes,no}   Is this a test run? (default: no)
  --n_years, -ny N_YEARS
  --scenario, -s {hist2,ssp370}

(cfa_env) sci-vm-01|Mon Mar 23|16:54:10 GMT|cfa_scripts|[main]>
```

Appendix C Priority variables

C.1 Background

The entirety of this report has dealt with the CF aggregation of the entire CANARI output as stored (currently, May 2026) on the JDMA system. For many applications however, only a subset of data is required and this is referred to as the ‘priority variables’ – PVs – set. Detailed information regarding the variables themselves can be found on the relevant [NCAS-CMS GitHub page](#).

C.2 Algorithm

For the priority variables, all data is accessible ‘live’ on the ‘canari’ Group Workspace on JASMIN at ‘/gws/ssde/j25b/canari/large-ensemble/priority-variables’. In addition, the PVs are pre-separated into single variable files, for example (see Figure 14):

- cv575a_1_1hr_m01s05i216.nc, for the atmosphere.
 - Surface temperature after timestep [°K], details of variables can be found on the [UM STASH browser](#).
- cv575o_1_day__grid_T_tos.nc, for the ocean.
 - Sea surface temperature [°C].
- cv575i_1_day_aice.nc, for the sea ice.
 - Ice area (aggregate) $\in [0, 1]$.

Because of the availability of the PVs on disk, the process for obtaining the CFA files representing each realm is greatly simplified and is performed by a single Python script,

`priority-variables.py`. This script reads in all the data for each member of the historical and SSP3-7.0 ensembles and writes out one CFA files for each variable/scenario/member combination. The `priority-variables` Python script which performs this is illustrated using PlantUML in Figure 13.

Listing 8: Output of `python priority-variables.py --help`

```
python priority-variables.py --help
Usage: priority-variables.py [OPTIONS]

    Create seed CFA file for CANARI priority variables.

Options:
  --realm [ATM|OCN|CICE]    The climate realm [required]
  --verbose INTEGER RANGE   Verbosity level of cf.read [-1<=x<=3]
  --scenario [HIST2|SSP370] The scenario [required]
  -m, --member INTEGER      The ensemble member number [required]
  -d, --data_path PATH       Path to the source data directory [required]
  --help                    Show this message and exit.
(cfa_env) sci-vm-01|Wed May 27|15:44:32 BST|priority-variables|[priority-variables]>
```

The code for generating Figure 14 is shown in Listing 9.

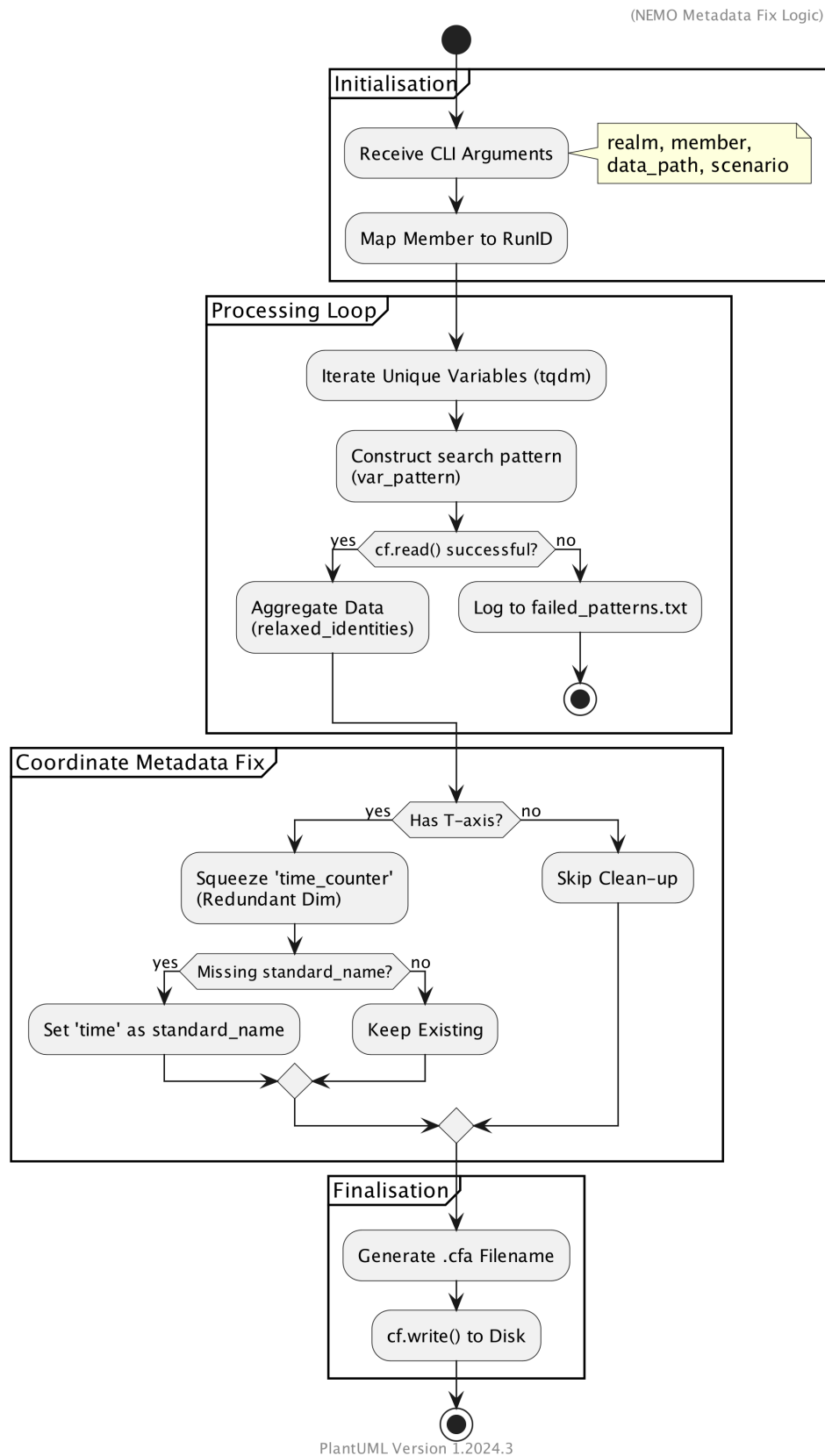


Figure 13: Algorithm for the `priority-files.py` file, Figure prepared using PlantUML.

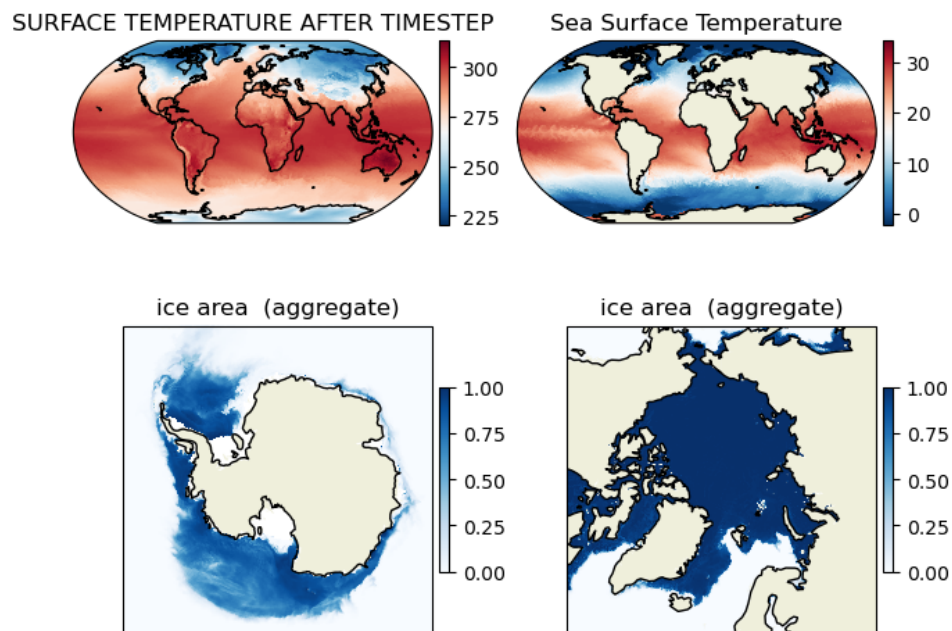


Figure 14: Example surface air temperature after timestep (in Kelvin), sea surface temperature (in Celsius), and sea ice area (aggregate) from ensemble member 1 of the historical ensemble.

Listing 9: Code to generate Figure 14.

```
import matplotlib.pyplot as plt
import cartopy.crs as ccrs
import cartopy.feature as cfeature
import cf

files = [
    "./CF-1.13_seed_CANARI_1_cv575_ATM_1_mon_m01s00i024_3.cfa",
    "./CF-1.13_seed_CANARI_1_cv575_OCN_day__grid_T_tos.cfa",
    "CF-1.13_seed_CANARI_1_cv575_CICE_day_aice.cfa",
] + [
    "CF-1.13_seed_CANARI_1_cv575_CICE_day_aice.cfa",
]

fig, axes = plt.subplots(
    nrows=2,
    ncols=2,
    layout="constrained",
)

if len(files) == 1:
    axes = [axes]
else:
    axes = axes.flatten()

cmaps = ["RdBu_r"] * 2 + ["Blues"] * 2

for i in range(len(files)):
    old_ax = axes[i]
    pos = old_ax.get_subplotspec()
    old_ax.remove()

    if i < 2:
        proj = ccrs.Robinson()
    elif i == 2:
        proj = ccrs.SouthPolarStereographic()
    elif i == 3:
        proj = ccrs.NorthPolarStereographic()

    ax = fig.add_subplot(pos, projection=proj)

    fields = cf.read(files[i])
    f = fields[0][0]

    mesh = ax.pcolormesh(
        f.construct("X").data.array,
        f.construct("Y").data.array,
        f.data.array.squeeze(),
        transform=ccrs.PlateCarree(),
        shading="auto",
        cmap=cmaps[i],
        zorder=0,
    )

    ax.coastlines(resolution="110m", linewidth=1)

    if i == 2:
        ax.set_extent([-180, 180, -90 * (i == 2), -60 * (i == 2)], ccrs.PlateCarree())
```

```
elif i == 3:
    ax.set_extent([-180, 180, 60, 90], ccrs.PlateCarree())
else:
    ax.set_global()

if i > 0:
    ax.add_feature(cfeature.LAND, zorder=1)

plt.colorbar(mesh, ax=ax, orientation="vertical", pad=0.02, shrink=0.6)

title_text = getattr(f, "long_name", f"File {i}")
ax.set_title(title_text)

plt.show();
```

C.3 Discovery and generation of single-variable files

The full CANARI data set is held on the JDMA system – see page 3 – but to retrieve individual variables there is a 2 step process:

1. Retrieve monthly data from JDMA for the years(s) required (see Figure 15).
2. Extract the required data and write to single-variable files (see Figure 16).

C.3.1 Retrieval

Figure 15 shows this graphically and Listing 10 gives the code implementation.

Listing 10: Output of `python jdma_get_monthly.py --help`

```
> python jdma_get_monthly.py --help
usage: jdma_get_monthly.py [-h] [--start START] [--end END] [--cycle CYCLE] [--
    filelist] [--ens ENS] suite

positional arguments:
  suite                Suite ID to extract data from

options:
  -h, --help          show this help message and exit
  --start START        Year from which to start extracting data from
  --end END            Year to extract data up to (included)
  --cycle CYCLE        Cycle (YYYYMM) for which to extract data for. Cannot be used in
                        combination with --start/--end
  --filelist           Create file(s) containing a list of files that would be extracted
  --ens ENS            Ensemble number of suite
(cfa_env) sci-vm-01|Tue May 26|15:27:35 BST|fix_corrupted|[priority-variables]>
```

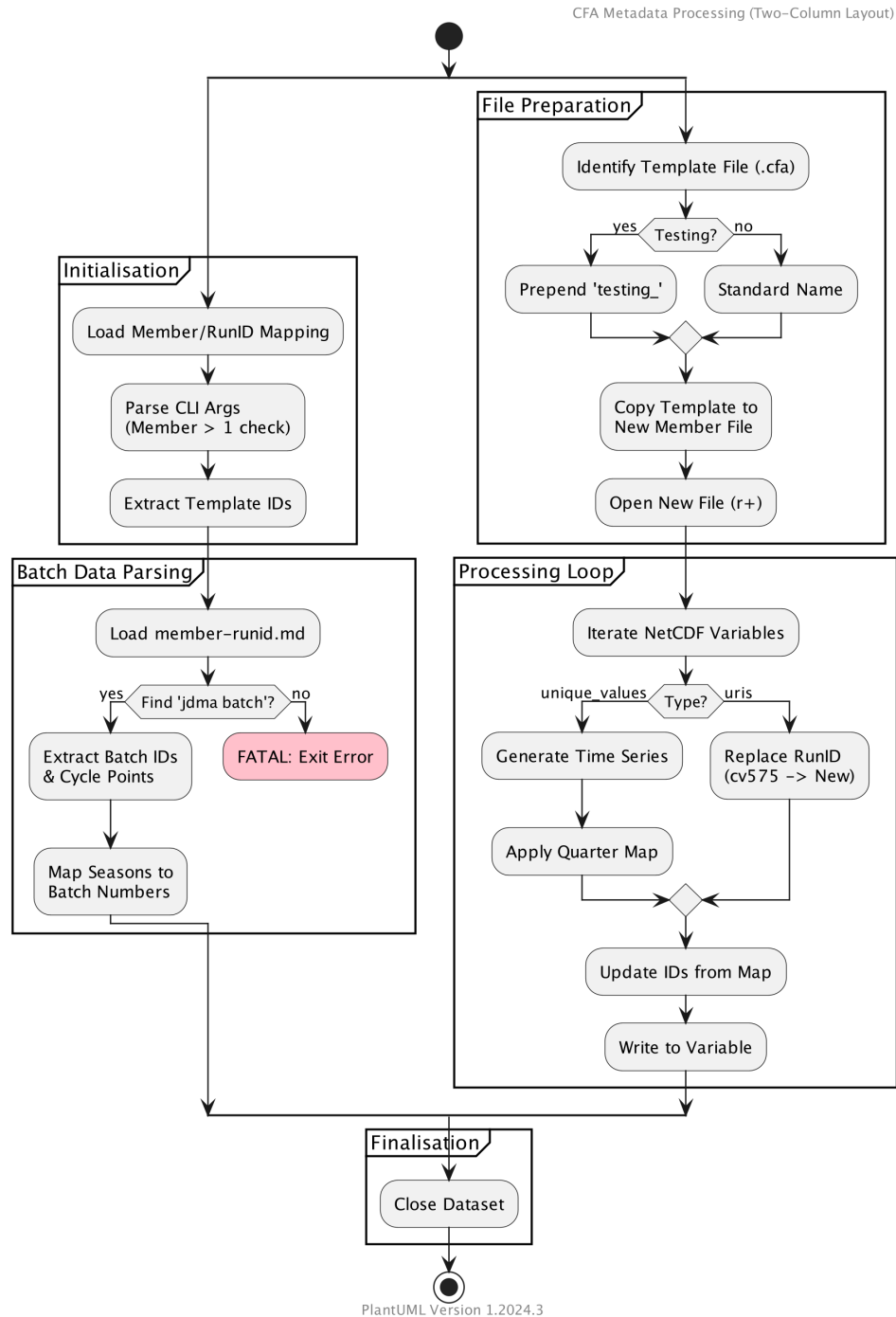


Figure 15: Algorithm for the `jdma_get_monthly.py` file, Figure prepared using PlantUML.

C.3.2 Extraction to yearly

Figure 16 shows this graphically and Listing 11 gives the code implementation.

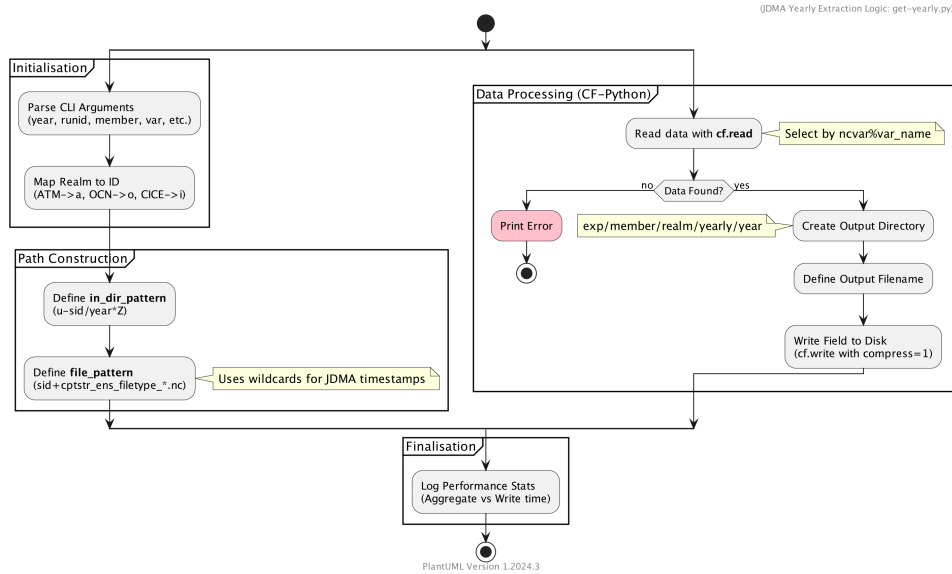


Figure 16: Algorithm for the `get-yearly.py` file, Figure prepared using PlantUML.

Listing 11: Output of `python jget-yearly.py --help`

```

> python get-yearly.py --help
usage: get-yearly.py [-h] --year YEAR --runid RUNID --member MEMBER --scenario
    SCENARIO --realm {ATM,OCN,CICE}
    --filetype FILETYPE --var VAR

Extract single-year variables from JDMA-retrieved data.

options:
  -h, --help            show this help message and exit
  --year YEAR           Year to process (e.g. 1975)
  --runid RUNID         Run ID without the u- prefix (e.g. cz649)
  --member MEMBER       Ensemble member number
  --scenario SCENARIO   Scenario (e.g. HIST2)
  --realm {ATM,OCN,CICE}
                        Realm
  --filetype FILETYPE   Type (e.g. mon__diaptr)
  --var VAR             Var name (e.g. zotemat1)
  
```

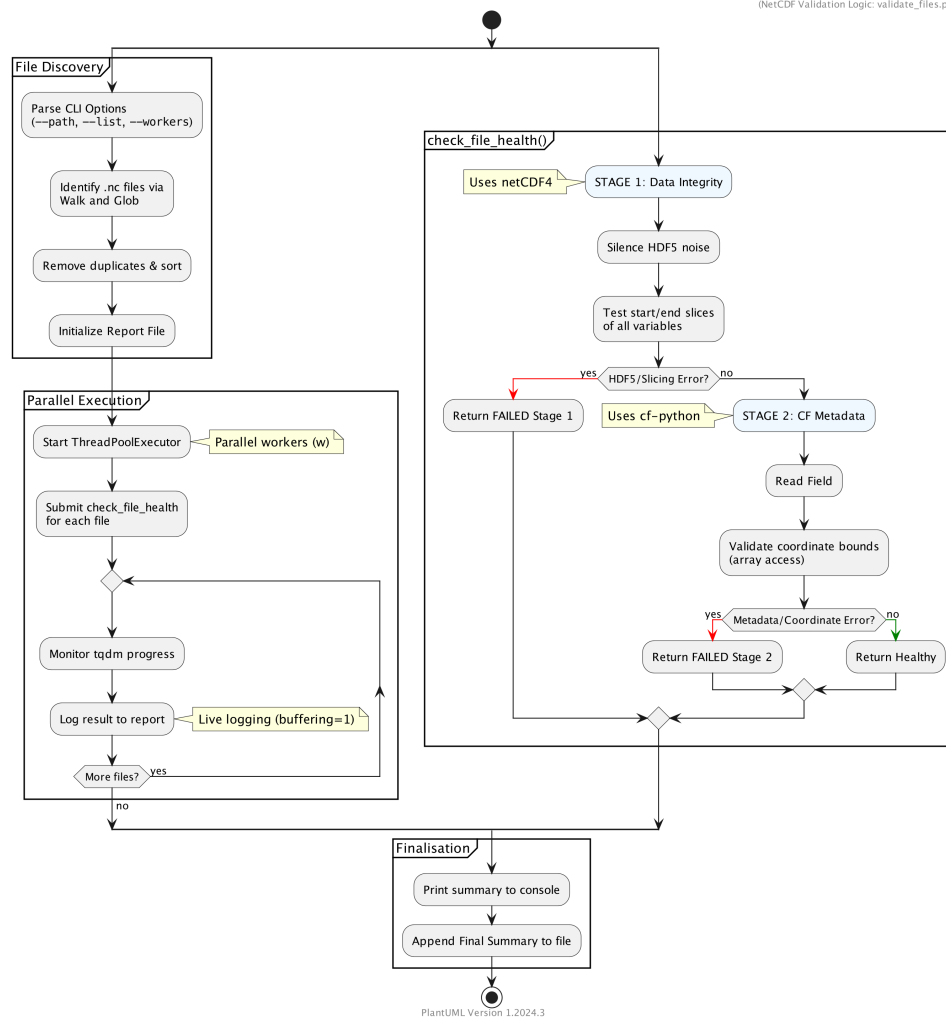


Figure 17: Algorithm for the `validate-files.py` file, Figure prepared using PlantUML.

C.4 Validation of corrupt priority-variable files

Figure 17 shows the validation process graphically and Listing 12 shows the code implementation.

Listing 12: Output of `python validate-files.py --help`

```

> python validate-files.py --help
Usage: validate-files.py [OPTIONS]

Scan NetCDF files and generate a full Health Report (Live Logging).

Options:
  -p, --path TEXT      Path to scan.
  -l, --list PATH      Text file with paths.
  -w, --workers INTEGER Number of parallel workers.
  -o, --out TEXT       Output report file.
  --help              Show this message and exit.
(cfa_env) sci-vm-01|Tue May 26|16:02:45 BST|fix_corrupted|[priority-variables]>
  
```

The validation process is twofold; the checking of the presence of coordinate bounds, and the fidelity of the first and last slices of each diagnostic. The choice of these two tests was heuristic in the sense that there are many different tests that could be performed to assess the quality of model data but these tests were found sufficient to detect – and subsequently fix – all corrupt

files.

Note that to enable corrupt-file discovery the `priority-variables.py` script shown in Figure 13 and Listing 8 enables the writing of Python ‘glob’ strings which cause the CFA generation process to fail. These lists can then be used to ‘home in’ on specific broken files.

References

- Brian Eaton, Jonathan Gregory, Bob Drach, Karl Taylor, Steve Hankin, John Caron, Rich Signell, Phil Bentley, Greg Rappa, Heinke Höck, Alison Pamment, Martin Juckes, Martin Raspaud, Jon Blower, Randy Horne, Timothy Whiteaker, David Blodgett, Charlie Zender, Daniel Lee, David Hassell, Alan D. Snow, Tobias Kölling, Dave Allured, Aleksandar Jelenak, Anders Meier Soerensen, Lucile Gaultier, Sylvain Herlédan, Fernando Manzano, Lars Bärring, Christopher Barker, Sadie L. Bartholomew, Thomas Lavergne, Bryan Lawrence, Neil Massey, Antonio S. Cofiño, Seth McGinnis, and Patrick Van Laake. NetCDF Climate and Forecast (CF) Metadata Conventions, December 2025. URL <https://doi.org/10.5281/zenodo.17801666>.
- D. Hassell, J. Gregory, J. Blower, B. N. Lawrence, and K. E. Taylor. A data model of the climate and forecast metadata conventions (cf-1.6) with a software implementation (cf-python v2.1). *Geoscientific Model Development*, 10(12):4619–4646, 2017. doi: 10.5194/gmd-10-4619-2017. URL <https://gmd.copernicus.org/articles/10/4619/2017/>.
- International Organization for Standardization. Date and time – Representations for information interchange – Part 1: Basic rules. ISO 8601-1:2019, 2019. URL <https://www.iso.org/standard/70631.html>. Standard.
- IPCC. Summary for policymakers. In *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. IPCC, Geneva, Switzerland, 2023. doi: 10.59327/IPCC/AR6-9789291691647.001.
- Hilary J. Oliver, Matthew Shin, and Oliver Sanders. Cylc: A workflow engine for cycling systems. *Journal of Open Source Software*, 3(27):737, 2018. doi: 10.21105/joss.00737. URL <https://doi.org/10.21105/joss.00737>.
- Arnaud Roques. Plantuml: Open-source tool that uses simple textual description to draw uml diagrams, 2026. URL <https://plantuml.com/>. Accessed: 2026-03-19.
- R. Schiemann et al. The CANARI HadGEM3 Large Ensemble: Design and evaluation of historical simulations. in prep., 2026.
- K. D. Williams, D. Copsey, E. W. Blockley, A. Bodas-Salcedo, D. Calvert, R. Comer, P. Davis, T. Graham, H. T. Hewitt, R. Hill, P. Hyder, S. Ineson, T. C. Johns, A. B. Keen, R. W. Lee, A. Megann, S. F. Milton, J. G. L. Rae, M. J. Roberts, A. A. Scaife, R. Schiemann, D. Storkey, L. Thorpe, I. G. Watterson, D. N. Walters, A. West, R. A. Wood, T. Woollings, and P. K. Xavier. The Met Office global coupled model 3.0 and 3.1 (GC3.0 and GC3.1) configurations. *Journal of Advances in Modeling Earth Systems*, 10(2):357–380, 2018. doi: 10.1002/2017MS001115. URL <https://doi.org/10.1002/2017MS001115>.